

LOAD BALANCER

Importance of load balancer:

- 1 Distributes incoming traffic evenly across multiple servers to prevent overload.
- 2 Ensures high availability by routing traffic only to healthy instances.
- 3 Increases fault tolerance by automatically handling instance or AZ failures.
- 4 Improves application performance and reduces latency.
- 5 Works with Auto Scaling to handle traffic spikes smoothly.
- 6 Enhances security with SSL termination, WAF, and DDoS protection.
- 7 Provides centralized monitoring and logging for better visibility and troubleshooting.

Tools going to be used for the task:

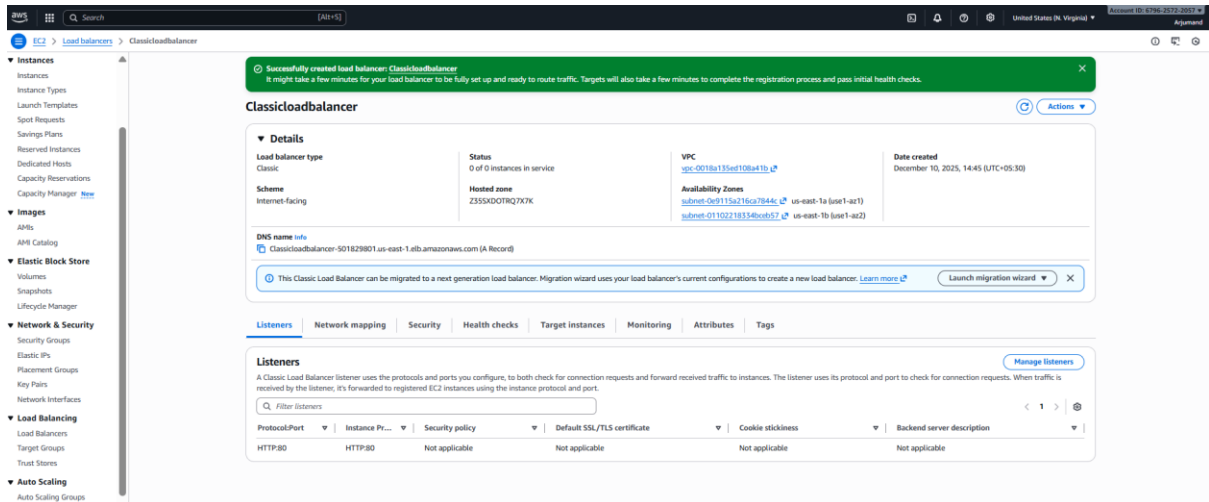
1. **EC2** → to launch instances for all load balancers.
2. **VPC (subnets + security groups)** → networking setup for CLB, ALB, NLB.
3. **Target Groups** → required for ALB and NLB routing.
4. **ACM (AWS Certificate Manager)** → to generate and attach SSL certificate to ALB.
5. **Route 53** → to map ALB DNS using an Alias record.
6. **S3 Bucket** → to store Application Load Balancer access logs.

1. Configure Classic Load balancer.

Steps:

- Go to EC2 → Load Balancers → Create Load Balancer → Classic Load Balancer.
- Choose Internet-facing, select public subnets, and keep listener HTTP:80.
- Configure security groups to allow inbound HTTP (port 80).
- Add health check using HTTP on path / and default thresholds.
- Register your EC2 instances, save, and verify they show InService.

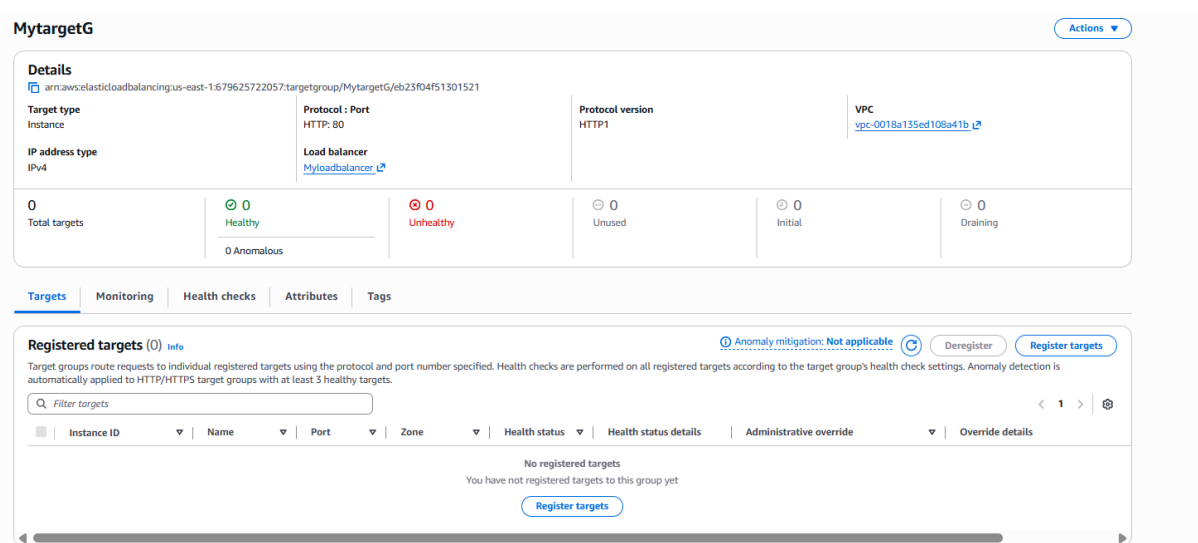
- I have configured classic load balancer with two public subnets in different zones



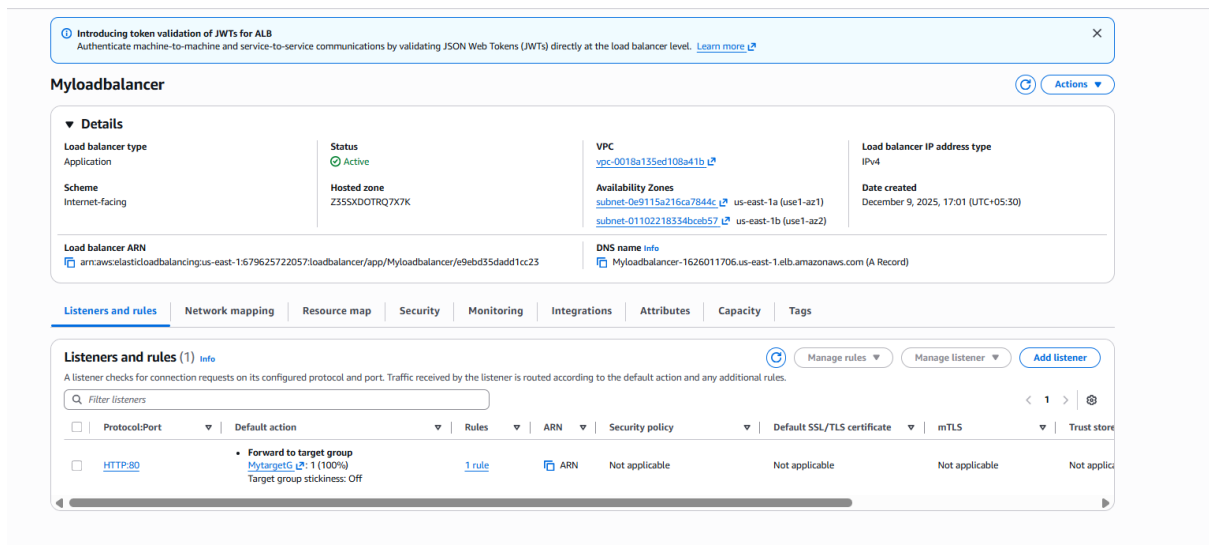
2. Configure Application Load balancer.

Steps:

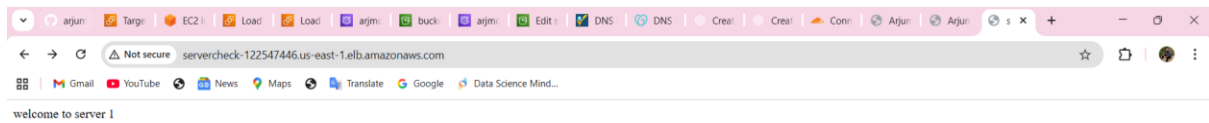
- Go to **EC2** → **Load Balancers** → **Create Load Balancer** → **Application Load Balancer**.
- Select **Internet-facing**, choose **IPv4**, and pick **at least two public subnets**.
- Create/choose a **Security Group** that allows inbound HTTP (80).
- Create a **Target Group** (Instance type → HTTP:80) and **register your EC2 instances**.
- In the ALB Listener section, add listener **HTTP:80** → **Forward to Target Group**.
- Create the ALB and verify that **Targets show “healthy”** and ALB DNS loads your website.



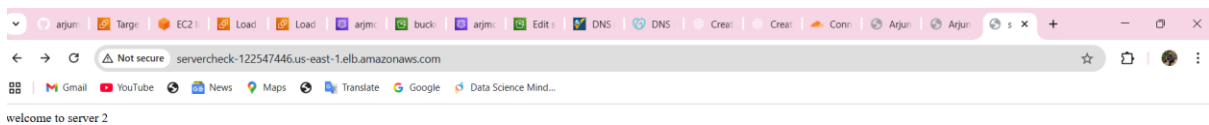
- Created ALB using two public subnets and one target group and security group



Load balancer working fine



When I refresh it is giving second load balancer



3. Configure Network Load balancer.

Steps:

- Go to EC2 → Load Balancers → Create Load Balancer → Network Load Balancer.
- Enter a name, choose Internet-facing, and keep TCP:80 as the listener.
- Select two public subnets across different Availability Zones.
- Create a new Target Group with Target Type = Instance and Protocol = TCP:80.
- Register your EC2 instances as targets and save the target group.
- Attach the target group to the NLB and click Create Load Balancer.
- Test using the NLB DNS name to confirm traffic reaches your backend EC2 instances.

REVIEW AND CREATE

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: [ALB](#)

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: [NLB](#)

Target group name
Name must be unique per Region per AWS account.

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 0/32

Protocol
Protocol for communication between the load balancer and targets.
TCP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0018a135ed108a41b (My_vpc_virginia) 12.0.0.0/18 [Create VPC](#)

EC2 > Target groups > Create target group

Step 1: Create target group
Step 2 - recommended: Register targets
Step 3: Review and create

Review and create
Review your target group configuration before creating

Step 1: Target group details [Edit](#)

Target group details			
Name NLBtarget	Target type Instance	Protocol : Port TCP: 80	Protocol version HTTP1
VPC vpc-0018a135ed108a41b ↗	IP address type IPv4		

Health check details			
Health check protocol HTTP	Health check path /	Health check port traffic-port	Interval 30 seconds
Timeout 6 seconds	Healthy threshold 5	Unhealthy threshold 2	Success codes 200-399

Step 2: Register targets [Edit](#)

Targets (0)

Instance ID	Name	Port	Zone
No targets added			

[Cancel](#) [Previous](#) [Create target group](#)

Created a new target group for NLB

Review
Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

Summary
Review and confirm your configurations: [Estimate cost](#)

Basic configuration Edit Name: mynlb Scheme: Internet-facing IP address type: IPv4	Network mapping Edit VPC: vpc-0018a135ed108a41b Availability Zones and subnets: - us-east-1a subnet-023fb7044fd931fb8 - us-east-1b subnet-01102218334bceb57	Security groups Edit default sg-09bf5507d74a26ce8	Listeners and routing Edit TCP:80 Forward to 1 target group
Service integrations Edit AWS Global Accelerator: -		Tags Edit -	
Attributes ⓘ Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.			

Creation workflow and status

► **Server-side tasks and status**
After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

[Cancel](#) [Create load balancer](#)

NLB created by new target group and two subnets

Successfully created load balancer

Introducing QUIC passthrough for NLB
Enable faster, more reliable mobile app connections with native QUIC protocol support, optimizing performance across varying network conditions. [Learn more](#)

mynlb [Actions](#)

Details

Load balancer type Network	Status ⓘ Provisioning	VPC vpc-0018a135ed108a41b	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZZ6RNL4JYFTOTI	Availability Zones subnet-023fb7044fd931fb8 us-east-1a (use1-az1) subnet-01102218334bceb57 us-east-1b (use1-az2)	Date created December 10, 2025, 15:25 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:679625722057:loadbalancer/net/mynlb/351754bce1d9f9fa		DNS name info mynlb-351754bce1d9f9fa.elb.us-east-1.amazonaws.com (A Record)	

[Listeners](#) [Network mapping](#) [Resource map](#) [Security](#) [Monitoring](#) [Integrations](#) [Attributes](#) [Capacity](#) [Tags](#)

Listeners (1) [Actions](#) [Add listener](#)

A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

☐	Protocol:Port	Default action	ARN	Security policy	Default SSL/TLS certificate	ALPN policy	Tags
☐	TCP:80	Forward to target group NLBtarget 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	None	0 tags

Validation load balancer working fine

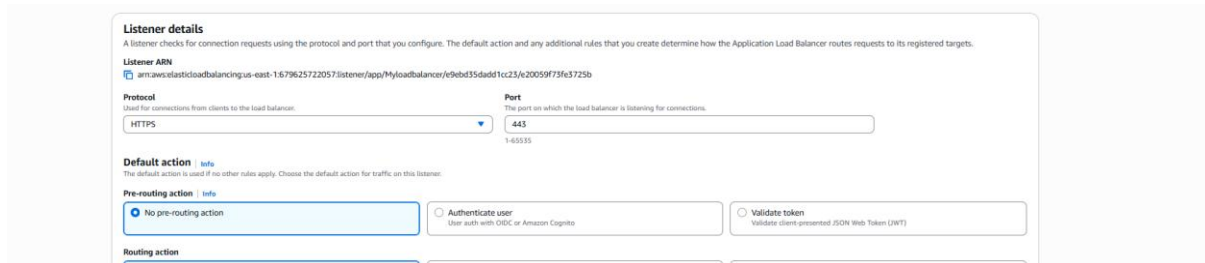


4. Attach SSL for application load balancer.

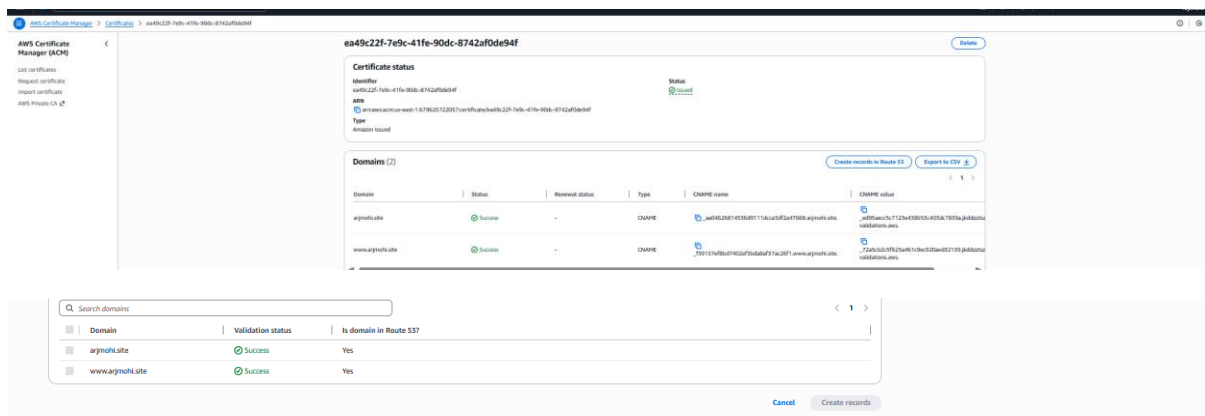
Steps:

- Go to EC2 → Load Balancers → Select your ALB.
-

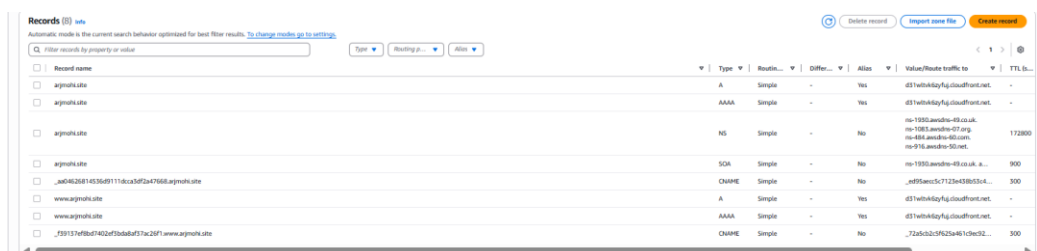
- Open the Listeners tab and click Edit on the HTTPS (443) listener.
-
- Under Secure listener settings, choose a Security policy (recommended TLS13/TLS12).
-
- Set Certificate source = From ACM.
-
- Select your issued certificate (example: arjmohi.site).
-
- Choose the target group you want to forward HTTPS traffic to.
-
- Click Save changes to enable SSL on the ALB.
- Created listener details http to https as we are attaching ssl certificate



- We can see the records after making certificate with my domain



- R53 records as we can see related to my domain



- Added ssl to my APLB and added certificate source

Pre-routing action | Info

☒ No pre-routing action

☐ Authenticate user
User auth with OIDC or Amazon Cognito

☐ Validate token
Validate client-presented JSON Web Token (JWT)

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group | Info

Choose a target group and specify routing weight or [create target group](#).

Target group

MytargetG
Target type: Instances, IPv4 | Target stickiness: Off

HTTP

Weight

1

0.999

Percent

100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness | Info

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Secure listener settings | Info

Security policy | Info

Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#)

Security category

All security policies

Policy name | [New](#)

ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate

The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source

☒ From ACM

☐ From IAM

☐ Import certificate

Certificate (from ACM)

The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.

arjmo4site
b07950df-acae-47ef-bdeb-5596d62d805c

[Request new ACM certificate](#)

Client certificate handling | Info

Client certificates are used to make authenticated requests to remote servers. [Learn more](#)

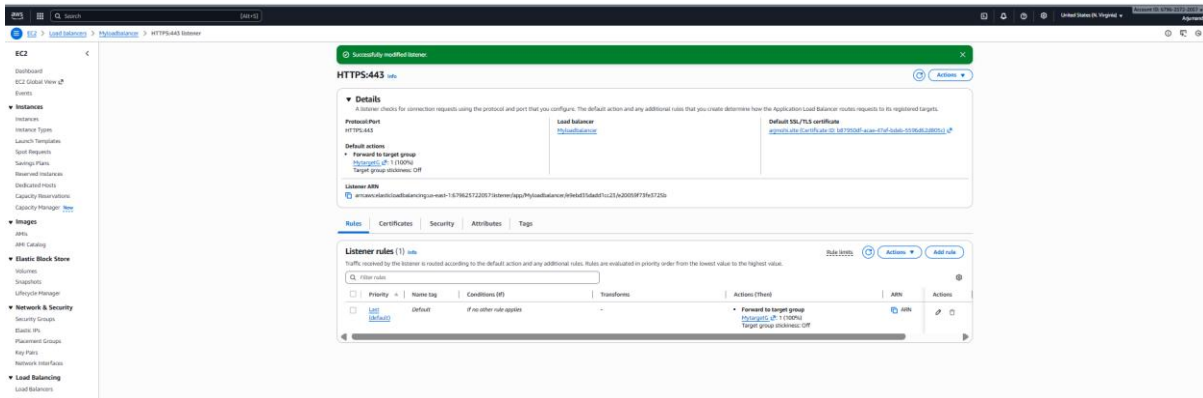
☐ Mutual authentication (mTLS)
Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

Cancel

Save changes



5. Map Application load balancer to R53.

Steps:

- Open Route53 hosted zone for your domain.
- Create an **A record** with Alias = **Yes**.
- Select “Alias to Application Load Balancer.”

- Choose your ALB from the drop-down list (dualstack).
 - Save the record and verify that the domain resolves to ALB.
- Mapped application load balancer to route 53

create record

Create record [info](#)

Quick create record

Record 1

Record name [info](#) Record type [info](#)

Keep blank to create a record for the root domain.

Alias ☒

Route traffic to [info](#)

US East (N. Virginia)

Alias hosted zone ID: Z33AQ2T9R727N

Routing policy [info](#)

Evaluate target health ☒

[Add another record](#)

[Cancel](#) [Create records](#)

View existing records

The following table lists the existing records in argmohs.site.

- New records are not added old records are deleted now records redirecting my load balancer

Hosted zone details [Edit hosted zone](#)

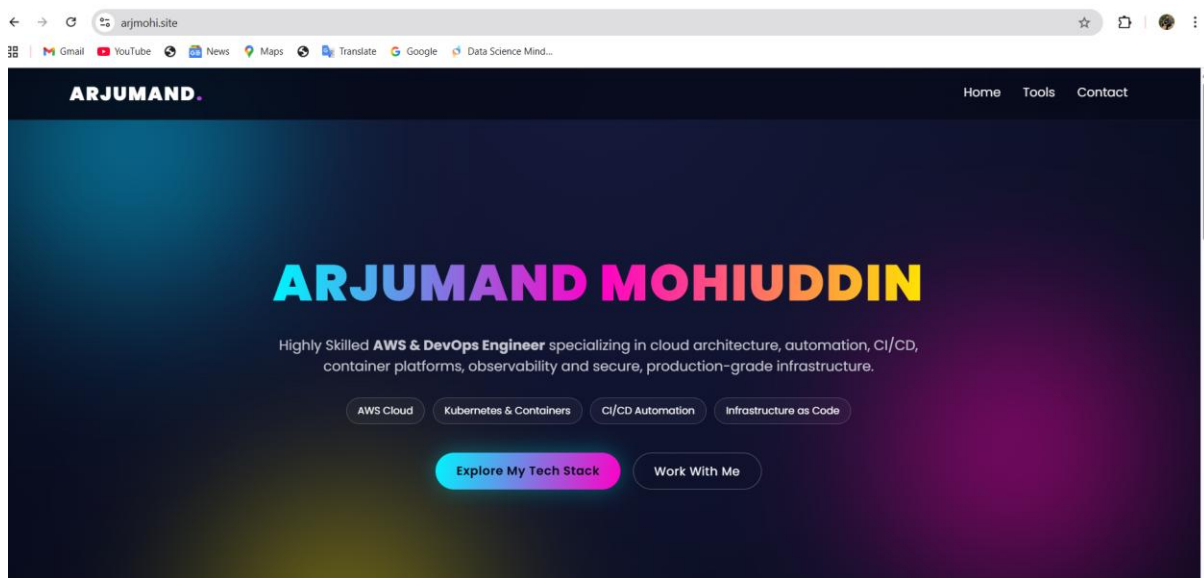
Records (6) [Accelerated recovery](#) [DNSSEC signing](#) [Hosted zone tags \(0\)](#)

Records (6) [info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

☐	Record name	Type	Routing...	Differ...	Alias	Value/Route traffic to	TTL (s)
☐	argmohs.site	A	Simple	-	Yes	dualstack-myloadbalancer-1...	-
☐	argmohs.site	AAAA	Simple	-	Yes	d37h6t48q7qj.cloudfront.net	-
☐	argmohs.site	NS	Simple	-	No	ns-1080.awdns-48.co.uk; ns-1081.awdns-07.org; ns-484.awdns-40.com; ns-916.awdns-50.net	172800
☐	argmohs.site	SOA	Simple	-	No	ns-1950.awdns-49.co.uk. a...	900
☐	_uad462681-6236d9111dca3f2a47868.argmohs.site	CNAME	Simple	-	No	_uad55aacc5c7123a438b53c4...	300
☐	www.argmohs.site	A	Simple	-	Yes	dualstack-myloadbalancer-1...	-

- This is my website is deployed after the new records



6. Push the application load balancer logs to S3.

Steps

- Created an S3 bucket in the same region as the ALB to store access logs.
 - Added a bucket policy allowing logdelivery.elb.amazonaws.com to write logs.
 - Opened ALB in EC2 → Load Balancers → Attributes tab.
 - Enabled "Access Logging" and selected the S3 bucket.
 - Added optional prefix for log organization.
 - Saved configuration to start log delivery.
 - Verified log files in S3 after 5–10 minutes.
-
- Created bucket then added bucket policy to receive load balancer logs

bucketforloadbalancer1 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#) [View analyzer for us-east-1](#)

Block public access (bucket settings)

[Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) [View](#)

Block all public access

 Off

► Individual Block Public Access settings for this bucket

Bucket policy

[Edit](#)[Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#) [View](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AWSALBLogging",
      "Effect": "Allow",
      "Principal": {
        "Service": "logdelivery.elasticloadbalancing.amazonaws.com"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::bucketforloadbalancer1/*",
      "Condition": {
        "StringEquals": {
          "s3:x-amz-acl": "bucket-owner-full-control"
        }
      }
    },
    {
      "Sid": "AWSALBLogging2",
      "Effect": "Allow",

```

[Copy](#)

Object Ownership

[Edit](#)

- Added bucket in load balancer in attributes tabs

Protection

☐ Deletion protection

To prevent your load balancer from being deleted accidentally, turn on deletion protection. If you turn on deletion protection, you must turn it off before you can delete the load balancer.

Monitoring

☒ Access logs

Access logs deliver detailed logs of all requests made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#) [View](#)

S3 URI

s3://bucketforloadbalancer1

[View](#)[Browse S3](#)

Format: s3://bucket/prefix

☐ Connection logs

Connection logs deliver detailed logs of all connections made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#) [View](#)

☐ Health check logs - new

Health check logs deliver detailed logs of health checks for all targets in your Elastic Load Balancer's target groups. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#) [View](#)

[Cancel](#)[Save changes](#)

- Bucket is added in load balancer successfully

Cross-zone load balancing

On

AWS zone-aware integration

Disabled

Protection

Deletion protection

Off

Monitoring

Access logs

S3 location: [bucketforloadbalancer1](#) [View](#)

Connection logs

Off

Health check logs - new

Off

- We can check load balancer logs are receiving in the s3 bucket

bucketforloadbalancer1 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (1)

[Refresh](#)[Copy S3 URI](#)[Copy URL](#)[Download](#)[Open](#)[Delete](#)[Actions](#)[Create folder](#)[Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) [View](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#) [View](#)

Find objects by prefix

☒ Show versions

< 1 >

[Settings](#)

☐	Name	Type	Last modified	Size	Storage class
☐	AWSLogs/	Folder	-	-	-